

Chapter 12 — Memory and Experience Ledger

Field	Value
Status	Revised after whole-book constitutional and cross-chapter audit
Version	0.3
Scope	Preserve every conversation, observation, event, action and outcome as immutable experience while managing active context and derived memory separately.
Constitutional basis	Decisions 4, 11, 24, 30, 32, 40, 42, 44, 48, 53
Owner review	No unresolved Owner decision after whole-book audit

1. Purpose

Preserve every conversation, observation, event, action and outcome as immutable experience while managing active context and derived memory separately.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Experience Ledger	Append-only Experience Objects and integrity chain
Context Manager	Working and short-term Active/Cooling/Dormant context states
Conversation Service	Conversation Objects, participants, purpose, decisions and open work
Archive Index	Searchable dormant experience and retention-class access

3. Core State and Records

Object	Required content
Experience Object	Identity, timestamp, receipt time, content reference, provenance, classification, causation and integrity.
Context Entry	Experience/knowledge reference, active state, relevance, recurrence, unresolved status and Owner priority.
Conversation Object	Purpose, state, participants, decisions, questions, dependencies and next actions.
Memory Derivation	Summary, pattern or knowledge link with source experiences and method.

4. Runtime Workflow

1. Append interaction or event once with identity, time, provenance, classification and causal context.
2. Place relevant references in working context for the active task.
3. Move short-term context through Active, Cooling and Dormant; default active horizon is 30 days but relevance may extend it.
4. Run reflection to link, summarize, generalize or promote while preserving raw experience.
5. Retrieve knowledge first, patterns second and raw experience when needed.
6. Apply visibility and retention policies; protected deletion may cryptographically erase payload access and remove derived copies but never remove the Experience Object or its tombstone from the ledger sequence.

5. Interfaces and Contracts

All commands, queries and events are typed, versioned, authenticated and correlated. Commands request mutation from the authoritative owner; queries declare consistency and privacy scope; immutable events record completed facts with identity, causation, authorization, provenance and integrity metadata.

6. Failure and Recovery

- Ledger append unavailable: do not claim durable completion; buffer only under declared safe policy.
- Corrupt projection: discard and rebuild from authoritative ledger entries.
- Privacy scope uncertain: withhold content and expose only permitted metadata.
- Pattern is disproven: revise derived knowledge; retain the supporting and contradicting experiences.

7. Constitutional Guarantees

- HAL never forgets experiences merely because active context expires.
- Expiration changes accessibility, not preservation.
- Derived memory never replaces its source experiences.
- Shared, personal and protected visibility are explicit and policy-enforced.

8. Prohibited Behaviors

- No component may silently broaden authority, reinterpret Owner intent, erase dissent, or treat a derived projection as authoritative state.
- No degraded dependency may silently change policy, identity, trust, authentication, verification or evidence requirements.
- No provider, model, node, Presence, secret, relationship or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records correlation, identity, policy/authority context, state versions, decisions, limitations, failures, recovery and outcomes. Verification is risk-scaled and produces durable evidence. Status reporting is derived from governed state rather than model assertion.

10. Source Alignment and Review

This chapter directly implements Decisions 4, 11, 24, 30, 32, 40, 42, 44, 48, 53. The v0.2 rewrite restores the source-specific objects, workflows and guarantees missing from v0.1. Final disposition is recorded in the separate source-alignment and constitutional-audit reports.

Whole-Book Audit Resolution

The following requirements supersede any ambiguous earlier wording in this chapter and are incorporated into this edition:

- The Conversation Service is the sole semantic owner of Conversation Objects. The Chapter 23 Thread Service maintains transport and delivery projections only.
- The Experience Ledger sequence is immutable. Protected deletion uses cryptographic erasure, deletes derived copies, and preserves a minimal non-sensitive tombstone.
- Removal of an Experience Object identity, sequence position, or tombstone is prohibited without a constitutional amendment.